03/11/2025

TEST NO 04

Class 10 Mathematics

Chapter: Polynomials | **Max Marks:** 40| **Time:** 45 minutes

---

SECTION A: 1-mark Questions

(Attempt all. Each question carries 1 mark)

1. What is the degree of the polynomial $3x^3 + 2x - 5$?
2. What do you call a polynomial of degree two?
3. Write the coefficient of x^2in $5x^2 - 4x + 3$.
4. State whether $f(x) = x^2 + 3x + 2$is a quadratic polynomial.
5. If $x = 1$is a zero of $x^2 - 6x + 5$, find the value of $k$in $x^2 - 6x + k = 0$.

SECTION B: 2-mark Questions

(Each question carries 2 marks)

6. Find the value of the polynomial $p(x) = x^2 - 3x + 2$at $x = 0$.
7. If one zero of the polynomial $x^2 + 3x + k$is 2, find the value of k.
8. Write the quadratic polynomial whose zeroes are 4 and -2.
9. If sum and product of zeroes are 5 and 6 respectively, write the quadratic polynomial.
10. State whether $2x^2 + \frac{3}{x} + 1$is a polynomial. Why or why not?

SECTION C: 3-mark Questions

(Each question carries 3 marks)

11. Find zeroes of $x^2 - 2x - 8$and verify the relationship between the zeroes and coefficients.
12. If the zeroes of the polynomial $f(x) = x^2 + px + 12$are 3 and -4, find the value of p.
13. Divide $x^3 - 7x + 6$by $x - 1$and write the quotient and remainder.
14. Write a cubic polynomial whose zeroes are 2, -1, and 3.
15. If a cubic polynomial has zeroes 1, 2, and 4, write the polynomial.

SECTION D: 4-mark Questions

(Each question carries 4 marks; attempt any 2)

16. Find the zeroes of $2x^2 - 8x + 6$and verify the sum and product with formula.
17. Divide $x^3 - 6x^2 + 11x - 6$by $x - 2$and verify the division algorithm.
18. If the sum and product of zeroes of a quadratic polynomial are 3 and 4, respectively, write the polynomial and find its zeroes.

SECTION E: 5-mark Question

(Attempt any 1 out of 2)

19.

a) Find all zeroes of the polynomial $x^3 - 4x^2 + 5x - 2$,

b) Form a quadratic polynomial using any two zeroes and

c) Verify relation between sum, product of zeroes and the coefficients for quadratic part.

OR

20.

a) Show that $x^2 + 5x + 6$has two real zeroes

b) Factorise it and

c) Verify the sum and product of zeroes with -b/a and c/a.